

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1001	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
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Date:	01/10/2026	Duration:	60 minutes

Code of Conduct

I T. Tarun (1924260001) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: T. Tarun

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

REAL-TIME WATER MANAGEMENT SYSTEM

1. Understand the Problem Statement

1.1 Explain the problem in your own words.

The Real-Time Water Management System is proposed to address the difficulties involved in monitoring and managing water resources through manual or delayed processes. Information about water level, consumption, flow rate, pressure, leakage, and distribution may not be available at the required time. This can lead to water wastage, unnoticed leakage, supply problems, inaccurate records, and delayed maintenance.

The proposed system uses IoT sensors, a centralized database, software applications, and a monitoring dashboard to collect and present water-related information in real time. It enables responsible users to identify abnormal conditions quickly and take suitable action.

1.2 Identify the objectives of the proposed system.

- Monitor water levels and water consumption in real time.

- Detect leakage and abnormal water usage at an early stage.
- Monitor water flow rate and pressure.
- Generate alerts when abnormal conditions are detected.
- Maintain accurate and centralized water management records.
- Improve water distribution and reduce unnecessary wastage.
- Support faster maintenance and decision-making.
- Provide a centralized platform for monitoring and reporting.

1.3 State the scope of the problem.

The scope of the system includes monitoring water level, flow rate, pressure, consumption, and leakage using connected sensors. The system collects sensor data, processes the information, stores it in a database, and displays it through a software dashboard.

The system can be used in residential areas, apartments, educational institutions, commercial buildings, industries, and other facilities where water needs to be monitored and managed efficiently.

The system also provides alerts, reports, historical data, and user management. Future versions can include advanced prediction, automated pump control, mobile applications, and integration with smart-city systems.

2. Identify Key Stakeholders and Challenges

2.1 Identify all the stakeholders involved in the system.

- Water Management Authorities: Monitor overall supply, distribution, and resource usage.
- System Administrators: Manage users, devices, configurations, alerts, and system access.
- Maintenance Personnel: Investigate sensor faults, leakage alerts, pressure problems, and equipment issues.
- Consumers or Residents: View relevant consumption information and receive service-related notifications.
- Facility Managers: Use reports and dashboards to plan water usage and maintenance.
- IoT/Sensor Operators: Install, configure, test, and maintain field devices.
- Software Development and Support Team: Develop, secure, maintain, and improve the application.

2.2 Explain the roles and responsibilities of each stakeholder.

Stakeholder	Role	Responsibilities
Water Management Authorities	Overall monitoring and management	Monitor water supply, distribution, consumption, leakage, and system performance.
System Administrators	System management	Manage users, sensors, access permissions, thresholds, and system configurations.
Maintenance Personnel	Maintenance and repair	Identify leakage, sensor faults, pressure problems, and equipment failures and take corrective action.
Consumers / Residents	Water usage monitoring	Monitor consumption, respond to notifications, and report abnormal water conditions.
Facility Managers	Facility-level management	Monitor usage, analyze reports, and plan maintenance activities.
IoT / Sensor Operators	Sensor management	Install, configure, calibrate, test, and maintain sensors.
Software Development Team	Software development	Develop, test, secure, update, and maintain the software system.

2.2 Discuss the major challenges and issues faced by the stakeholders.

- Delayed identification of leakage and abnormal consumption.
- Dependence on manual readings and fragmented records.
- Large volumes of sensor data requiring continuous processing.
- Unreliable communication or temporary sensor disconnection.
- Difficulty distinguishing genuine abnormal events from faulty sensor readings.
- Need for secure access to operational and consumer information.
- Maintenance costs associated with distributed devices and communication infrastructure.
- Requirement for a system that can scale when new locations and sensors are added.

3. Analyze the Current Approach

3.1 Describe the existing system or current process.

The existing water management process generally depends on manual meter readings, periodic inspection, operator observation, and scheduled maintenance.

Water levels and consumption may be checked at specific intervals instead of continuously. When leakage or equipment failure occurs, the problem may only be identified after physical inspection or after a user reports it.

Reports may also be maintained manually, making it difficult to obtain accurate real-time information about the overall water system.

3.2 Identify its strengths and limitations.

Strengths

- Simple to understand and operate.
- Suitable for small-scale monitoring.
- Physical inspection can identify certain equipment problems.
- Does not require a complex software platform for basic operations.

Limitations

- Monitoring is not continuously available in real time.
- Manual data collection requires time and effort.
- Leakage may remain unnoticed for longer periods.
- Manual records can contain errors.
- Information may be distributed across separate records.
- Immediate automated alerts may not be available.
- Large-scale monitoring becomes difficult.

3.3 Analyze the problems or gaps that need to be addressed.

The major problems and gaps include:

1. Lack of real-time monitoring.
2. Delayed detection of leakage.
3. Manual data collection.
4. Possibility of human errors.
5. Lack of automatic alerts.
6. Difficulty in monitoring multiple locations.
7. Limited historical analysis.
8. Slow maintenance response.

9. Lack of centralized data.

10. Difficulty in identifying abnormal consumption.

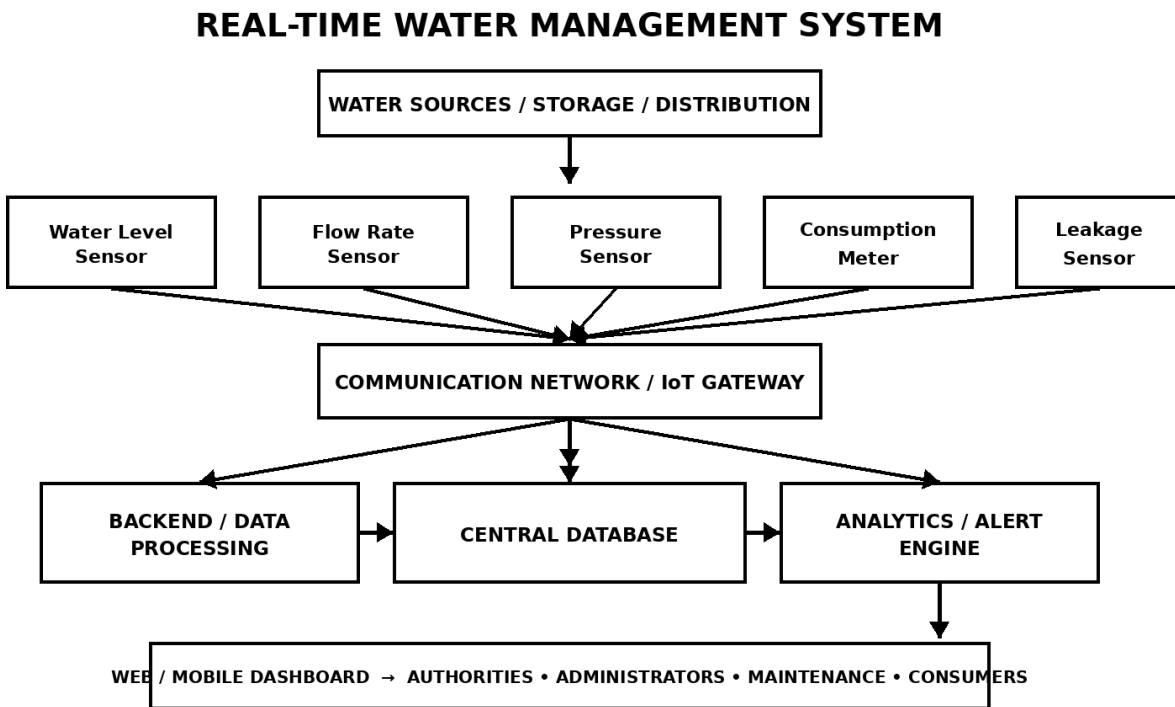
4. Propose a Suitable Solution Using Software Engineering Principles

4.1 Design a software-based solution for the given problem.

The proposed Real-Time Water Management System consists of water-level sensors, flow sensors, pressure sensors, and consumption monitoring devices.

These sensors collect water-related information and send the data through an IoT gateway or communication network. The backend system receives the data, processes it, and stores it in a database.

A web-based dashboard allows authorized users to monitor the current water status, view historical information, receive alerts, and generate reports.



4.2 Describe the major functional modules of the proposed system.

The major functional modules of the system are:

1. Sensor Data Collection Module

This module collects information from water-level, flow, pressure, and consumption sensors.

2. Real-Time Monitoring Module

This module displays the current water-related information to authorized users.

3. Water Consumption Monitoring Module

This module records and monitors water consumption over different time periods.

4. Leakage and Anomaly Detection Module

This module identifies unusual readings that may indicate leakage or abnormal water usage.

5. Alert and Notification Module

This module generates notifications when leakage, low water level, high consumption, or other abnormal conditions are detected.

6. User and Access Management Module

This module manages users and controls access to different system features.

7. Dashboard and Visualization Module

This module displays water information through charts, tables, indicators, and status information.

8. Reporting and Historical Analysis Module

This module generates reports and allows users to analyze previous water consumption data.

9. Device Management Module

This module monitors the status of connected sensors and devices.

4.3 Identify the functional and non-functional requirements.

Functional Requirements

- Collect data from water monitoring sensors.
- Monitor water level, consumption, flow rate, and pressure.
- Identify abnormal water usage and possible leakage.
- Generate alerts and notifications.
- Store historical water data.
- Provide dashboards and reports.
- Manage users and access permissions.

Non-Functional Requirements

- Reliability – the system should provide dependable monitoring.
- Performance – sensor information should be processed efficiently.
- Scalability – additional sensors and locations should be supported.
- Security – user and system information should be protected.
- Usability – the interface should be simple to understand.
- Maintainability – software components should be easy to update.
- Availability – the monitoring platform should be accessible when required.
- Accuracy – collected and reported information should be as accurate as possible.

4.4 Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.

- Modularity: The system is divided into independent modules such as monitoring, users, alerts, reporting, and sensor management.
- Scalability: New sensors, users, and locations can be added without redesigning the complete system.
- Maintainability: A modular structure makes testing, troubleshooting, and updating easier.
- Reliability: Continuous monitoring, data validation, and alerts improve dependable operation.
- Usability: A clear dashboard presents important water information in an understandable form.
- Security: Authentication, authorization, and protected data handling help prevent unauthorized access.

5. Justify the Chosen Methodology/Framework

5.1 Select an appropriate Software Development Life Cycle (SDLC) model or software development framework.

The Agile Software Development methodology using Scrum practices is selected for the Real-Time Water Management System.

The system contains several modules and may require changes based on feedback from stakeholders. Agile allows the system to be developed in smaller iterations instead of developing the entire system at once.

5.2 Justify why the selected methodology/framework is suitable for the proposed system.

Agile is suitable because the requirements of the water management system may change during development.

The project can be divided into different development stages such as sensor integration, database development, dashboard development, alert management, reporting, and user management.

Stakeholders can provide feedback after each iteration, and the development team can make improvements based on that feedback.

5.3 Explain how it supports successful project development and maintenance.

- Supports incremental development and frequent testing.
- Allows stakeholder feedback to be included during iterations.
- Makes it easier to change or add requirements.

- Helps identify defects earlier through continuous testing.
- Supports gradual addition of new sensors and software features.
- Makes future maintenance and enhancement more manageable.

6. Discuss Expected Outcomes and Limitations

6.1 Describe the expected benefits of the proposed solution.

- Real-time monitoring of water levels and consumption.
- Faster identification of possible leakage.
- Reduction in unnecessary water wastage.
- Improved water distribution management.
- Automatic alerts for abnormal conditions.
- Accurate and centralized water records.
- Improved maintenance planning.
- Better decision-making using timely information.
- Improved visibility of water usage across locations.
- More efficient overall water resource management.

6.2 Explain how the solution addresses the identified challenges.

The proposed system addresses the identified challenges by providing continuous monitoring through sensors.

Instead of depending completely on manual inspection, the system automatically collects and processes water-related information.

Automated alerts help maintenance personnel identify leakage and abnormal conditions quickly. The centralized database allows authorized users to access current and historical information from one platform.

The dashboard and reports also help facility managers and water management authorities make better decisions.

6.3 Discuss potential risks, implementation challenges, and limitations.

The possible risks and implementation challenges include:

- Sensor failure or inaccurate sensor readings.
- Network and communication failures.
- Power interruptions.
- Cybersecurity threats.

- High initial installation cost.
- Regular sensor calibration and maintenance requirements.
- Database or server failures.
- Difficulty in integrating with existing infrastructure.
- Requirement for trained personnel.
- Connectivity problems in remote locations.

6.4 Suggest possible future enhancements.

Possible future enhancements include:

- Machine-learning-based leakage prediction.
- Predictive water demand analysis.
- Mobile application support.
- Automated pump and valve control.
- Smart-city platform integration.
- Advanced water consumption forecasting.
- Geographical map-based monitoring.
- Automated maintenance scheduling.
- Integration with weather and environmental information.
- Advanced analytics and visualization.
- Cloud-based monitoring.
- AI-based water management recommendations.

7. conclusion.

The Real-Time Water Management System provides a software-based approach for improving the monitoring and management of water resources. By combining sensors, IoT communication, a centralized database, automated alerts, and a user-friendly dashboard, the system can provide timely information about water consumption, leakage, water levels, and other important parameters.

The use of software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security helps create a system that can be developed and maintained effectively. The Agile methodology also supports iterative development and continuous improvement based on stakeholder feedback.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification (10%)	Selects the most suitable SDLC/model/framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

Criteria	Mark
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25